Project

Project scope and plan

Team 11

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1. Background/context and problem statement:

Study Group, a leader in international education, partners with universities to deliver tailored academic programs through its International Study Centres (ISCs). Study Group's goal is to empower international students to succeed at universities from across the UK, Ireland and the US[1].

Scholarships and discounts are critical tools for attracting a diverse student base, achieving enrolment targets, and fostering global inclusivity. However, the inconsistent allocation of these discounts across regions, programmes, and agents creates inefficiencies and missed opportunities.

The primary focus of this project lies in maximising the return on investment (ROI) of these discounts while minimising underutilisation. Study Group faces difficulties in identifying which discounts drive enrolments, understanding allocation trends, and addressing underperforming areas. These inefficiencies impact the organisation's ability to place the desired number of students into universities.

This project aims to address several core questions:

- *How effectively are discounts distributed across regions, programmes, and agents?*
- *Which discount types yield the highest enrolment and progression rates?*
- *Where are there gaps or missed opportunities in discount allocation?*
- *What demographic or external factors correlate with successful student enrolment?*

By analysing historical data and addressing these questions, the project will provide actionable insights to refine scholarship strategies, improve decision-making, and enhance Study Group's ability to support its student population.

[1] *Educational bridging solutions: Our solutions: Study group* (no date) *Our solutions | Study Group*. Available at: https://www.studygroup.com/our-solutions/educational-bridging-solutions (Accessed: 18 January 2025).

2. Team roles and ways of working

Team

Our team consists of four members, each bringing expertise in data analytics and visualisation. Roles are distributed based on individual strengths and preferences to ensure balanced collaboration and efficiency:

- **Bev Grantham** was identified as the Project Manager of the team and will leverage organisation and coordination skills to ensure the team meets all milestones.
- **Davide Casarin** will act as the Data Analysis Lead, guiding the team's efforts in data cleaning, exploratory data analysis (EDA), and advanced analytics. Whilst they lead in this area, all members will actively participate in data-related tasks.
- **Jessy Pardi** will serve as the Visualisation lead, creating impactful charts, dashboards, and presentations to clearly communicate our findings.
- **Emmanuel Wagaba** will focus on Quality Assurance, reviewing all deliverables to ensure accuracy and consistency.

Work Distribution

The team will collaborate on all phases of the project, especially in data analysis, and tasks will be divided fairly. The Data Analysis Lead will assign focus areas, and all members will contribute insights during weekly reviews. The Project Manager will oversee progress and ensure deadlines are met. Visualisation tasks will integrate findings from the entire team, with deliverables undergoing final quality assurance checks by the Quality Assurance lead.

Communication Plan

The team will communicate and collaborate using various platforms. WhatsApp will be used for daily communication and quick updates, while project meetings will be conducted via Google Meet for more in-depth discussions. Collaborative document sharing will be facilitated through Google Docs, allowing real-time input and revisions.

3. Project plan

The Study Group project will span five key phases delivered over 6 weeks:

1. **Planning and Scoping (Week 1):** The project begins with defining objectives, success criteria, scope, and team responsibilities. Deliverables include a project charter and a clear project plan and roadmap. These steps ensure that all team members are aligned.
2. **Data Preparation (Week 2):** In this phase, both internal and external datasets are gathered, cleaned, and integrated. Tasks include handling missing values, standardising formats, and merging datasets. The key deliverable is a clean dataset to be used in the wider analysis.
3. **Exploratory Data Analysis (EDA) (Week 2):** This phase focuses on identifying patterns, trends, and anomalies within the data through summary statistics and visualisations. Visualisations will be developed to communicate key findings and highlight areas requiring deeper investigation.
4. **Advanced Analysis and Modelling (Weeks 2, 3 & 4):** During this phase, statistical and advanced analytics techniques are applied to evaluate discount allocation, effectiveness, and demographic trends. Models will be applied to uncover relationships between discounts and enrolment rates and identify underutilised opportunities.
5. **Insights, Recommendations, and Reporting (Weeks 4, 5 & 6):** The final phase involves consolidating findings into a comprehensive presentation, with actionable recommendations to optimise Study Group's discount policies. The team will prepare two pitch presentations for Study Group, one mid-project to gather feedback and a final presentation to deliver refined insights.

A detailed project roadmap expanding on each of these phases can be found in Appendix A: Project Roadmap.

4. Analytical approach

The team will adopt a streamlined analytical process focused on efficiency and clarity. For data cleaning and preparation, Python will be used for its robust libraries, such as Pandas for data manipulation and NumPy for numerical operations.

The analysis will follow a clear timeline as highlighted in the Project Roadmap. Key milestones include completing data cleaning by the end of Week 2, completing exploratory analysis and using visualisations generated with Tableau, Matplotlib and Seaborn to identify key patterns and trends by the middle of Week 3, and finalising models and findings by the end of Week 5. Advanced analytics will be performed through regression models and clustering algorithms, for instance using Scikit-learn to evaluate the impact of discount strategies.

Tasks will be divided based on each member's strengths, with clear ownership assigned to different phases of the analysis. The use of shared Jupyter Notebook files will allow team members to review, comment on, and refine each other's work.

5. Risks

The risks relating to the project are documented below and will be regularly assessed to ensure any disruption to the project is minimised.

Risk	Risk Severity	Mitigations
Data quality: Data may be inconsistent, incomplete or inaccurate which could impact the analysis	Medium	• Thorough cleaning and validation of the data. • Engage stakeholders to verify data accuracy if issues are identified.
Team coordination: Working across different time zones and with different working hours may impact our ability to collaborate effectively as team	High	• Identify lead person(s) for each task early, ensuring equitable workload distribution. • Build buffer time into the project roadmap to account for unforeseen delays.

		• Maintain WhatsApp and Google Docs to enable real-time updates. • Escalate and address concerns early.
Overreliance on Models: Models may not fully capture the complexity of enrolment behaviour or external factors, which may lead to misleading insights.	Medium	• Quality Assurance Lead to review scripts to ensure models are applied appropriately. • Validate models using cross-validation techniques to ensure robustness. • Communicate limitations of the analysis in the final report and recommend areas for future study.

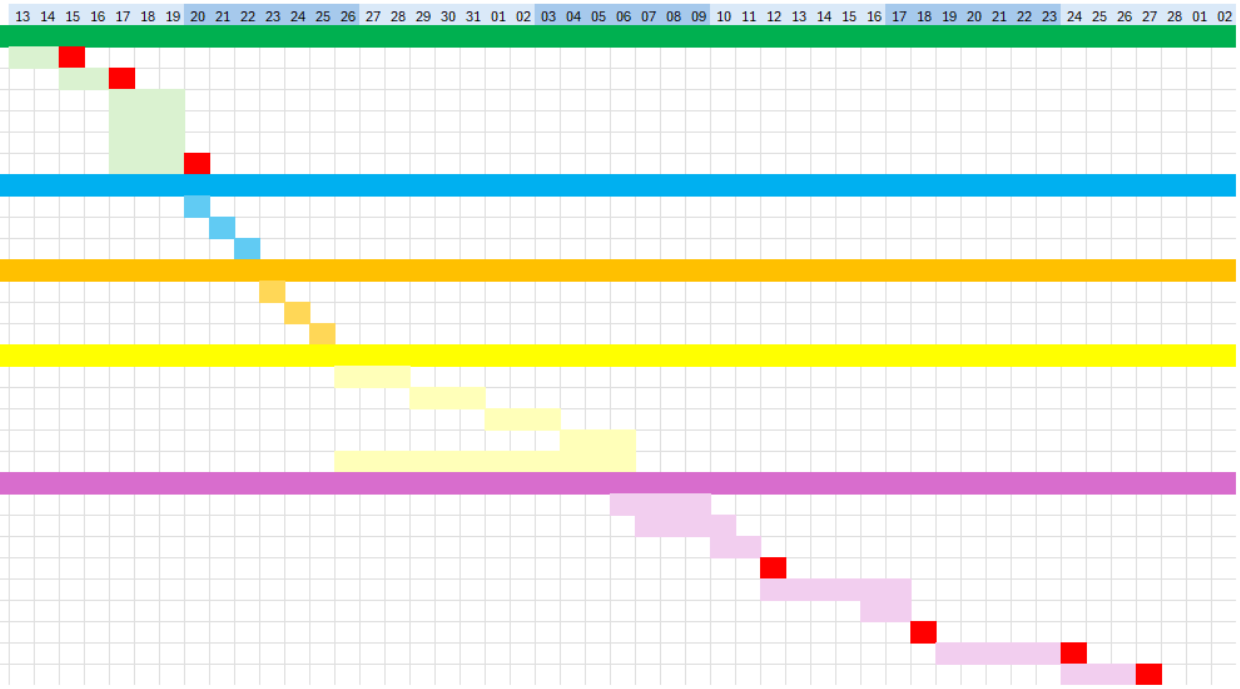
6. Conclusion

By addressing the challenges of discount allocation, this project seeks to provide Study Group with data-driven insights that enhance decision-making and strategic planning. Through a collaborative and methodical approach, the team will produce recommendations which can optimise Study Group's scholarship strategies, improve enrolment outcomes, and support their mission of empowering a diverse and inclusive global student community to succeed at university.

7.Appendix A: Project Roadmap

Key: Deadlines

Project plan and scope	Responsible person(s)	
Submission of non-disclosure agreements	All	☑
Team charter	All	☑
Identify project objectives	All	☑
Define project scope	All	☑
Assign team roles and responsibilities	All	☑
Develop project plan	Bev Grantham	☑
Data preparation		
Review Study Group dataset	All	☐
Clean and pre-process the data	Davide Cascarin	☐
Consider use of external data	Emmanuel Wagaba	☐
Exploratory analysis		
Generate summary statistics	Bev Grantham	☐
Identify anomalies and outliers	Jessy Pardi	☐
Develop visualisations of links between variables	Emmanuel Wagaba	☐
Analysis and modelling		
Analyse discount allocation	Davide Cascarin	☐
Evaluate discount effectiveness	Jessy Pardi	☐
Perform demographic and trend analysis	Bev Grantham	☐
Identify gaps and missed opportunities	Davide Cascarin	☐
Quality Assurance	Emmanuel Wagaba	☐
Insights, Recommendations, and Reporting		
Develop actionable recommendations to optimize discount allocation.	All	☐
Prepare a final presentation for stakeholders	Jessy Pardi	☐
Pitch rehearsal	All	☐
Initial pitch	All	☐
Interim review	All	☐
Final pitch rehearsal	All	☐
Final pitch	All	☐
Final document submission	Bev Grantham	☐
Individual reflections	All	☐



Percentage completion

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